

Spectral Pileup Correction and Noise Reduction Using CUDA Multi-Core Parallel Processing*

Zikang Chen,^{1,2} Xiaoying Zheng,^{1,2,†} Yongxin Zhu,^{1,2,‡} Tom Trigano,³ Yiwei Huang,^{1,2} and Yue Zhang¹

¹Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, China

²University of Chinese Academy of Sciences, Beijing 101408, China

³SCE, Department of Electrical Engineering, Ashdod Campus 77245, Israel

Spectroscopy devices suffer from the pulse pileup phenomenon, caused by overlapping of the signals. The energy-domain based pileup correction algorithm estimates the pulse energy distribution by measuring the duration and energies of pileups directly and does not need to identify each individual pulses. The correction algorithm can efficiently recovers the energy spectrum even under a very high photon arrival rate. However, the correction algorithm is sequential in nature and is slow when the energy resolution is high. A fast parallel implementation of the original correction algorithm is proposed in this paper. The parallel counterpart leverages state-of-the-art many-core system technology and achieves a nearly linear acceleration when the problem size scales. The speedup ratio exceeds 200 when the energy spectrum is split into 2,048 bins. Secondly, a low complexity Gaussian noise elimination method is introduced to work on the energy histogram directly and achieves better energy resolution when Gaussian noise exists. Finally, the algorithm performance limit under high activity is asymptotically analyzed.

Keywords: gamma, pile up, correction, parallel

I. INTRODUCTION

Gamma spectroscopy is present in many modern applications, ranging from astrophysics to medical imaging, and aims to analyze and use radioactive sources to probe material properties. In a typical gamma spectroscopy system, after the energy associated to portions of signals has been estimated, a spectrum analyzer builds a histogram of the computed energies, producing a spectrum similar to the one appearing in Fig. 1. In practice, the main perturbations which cripple the estimation of the terms of interest are not related to poor electronics, but are due to the stochastic nature of the recorded signal. A perturbation called the pileup effect stems from the random nature of the recorded electrical activity. In practice, the inter-arrival times between photons can be shorter than the durations of generated electrical pulses after amplification, thus creating clusters of pulses, known as *pileups* whose parametric characteristics have also been studied in numerous papers[1, 2]. An example of pileups is presented in Fig. 2. This also causes a distortion of the energy spectrum that becomes more severe as the activity increases. More specifically, standard spectrum analyzers consider pileups as single electrical pulses, and record the sum of the pulses individual energies instead of each energy separately. Therefore, fake sum peaks in the energy spectrum can be observed, which

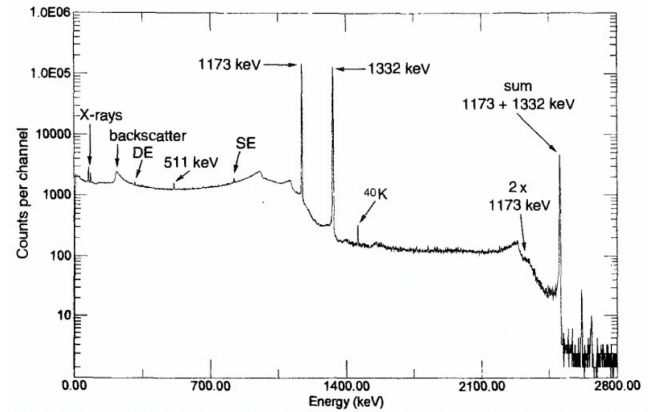


Fig. 1. Typical energy spectrum obtained from an isotope of Cobalt. Only the peaks appearing at 1173 and 1332 keV are representative of the Cobalt, while the continuum and the additional peaks are physical perturbations (from [13]).

make the final identification of the source more difficult for the practitioner. This has led to extensive research on algorithms designed to correct these artifacts [3–11].

There are the few known general ways to cope with the pileup effect referred as pileup correction algorithm[14, 15]. These approaches can be categorized into time-domain based and energy-domain based methods [16]. The most common time-domain approach is rejection. More specifically, we detect when pileup occurs and collectively reject all the pulses which took part in that incident. Such an approach can increase the duration of experiments, and some methods extend further this approach by compensating for the pulses which were rejected with the motivation to align the actual activity of the recorded signal and the activity of the processed, pileup effect clean signal. Other methods try to correct the estimate of the activity by introducing a correcting term in the activity estimator to take into account [17]. More recently it was sug-

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† Corresponding author, Xiaoying Zheng, Haik Road 99, Pudong New Area, Shanghai, 021-20350818, zhengxy@sari.ac.cn

‡ Corresponding authors: Yongxin Zhu, Haik Road 99, Pudong New Area, Shanghai, 021-20608151, zhuyongxin@sari.ac.cn.

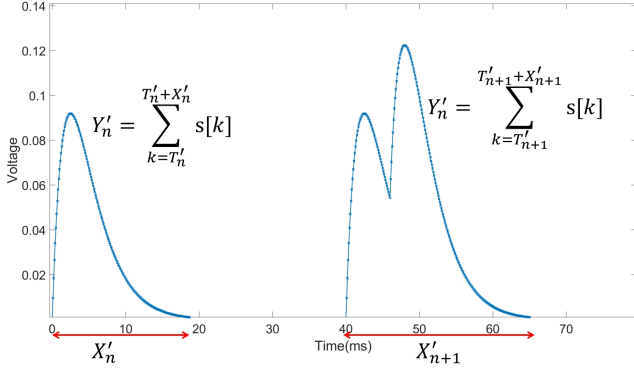


Fig. 2. pileup example

gested to consider the output signal as a linear combination of known shapes and to relate the pileup effect to a sparse regression problem [18–21]. Though efficient, these approaches are limited when the activity of the radioactive source increases too much. At a more general level, the importance of finding other methods going beyond signal processing of the signal in the time domain was also noted in [22]. On the other hand, energy-domain based strategies do not seek to identify and characterize individual pulses, but rather the pulses energy spectrum by means of statistical methods [16, 23–25]. Energy-domain based strategies circumvent the need to identify individual pulses, but still require considerable computations and do not always satisfy the requirement of real-time processing.

We consider in this paper the acceleration of pileup correction algorithm developed in [23]. This recursive correction algorithm estimates the pulse energy histogram from measurements of the duration and energies of overlapping pulses. Compared with time-domain algorithms, this approach can efficiently eliminate the pileup effect even under a high photon activity. Furthermore, its lower algorithmic complexity makes it more fitting for real-time implementation compared with the method presented in [25]. However, due to its sequential nature, the algorithm is still slow when the number of energy histogram bins is large and the energy resolution is high. In order to cope with this difficulty, we firstly propose to deploy the NVIDIA GPU (Graphics Processing Unit) technology to accelerate the pileup correction algorithm. GPUs are specialized for compute-intensive and massively parallel computations, and have become the *de facto* solution for parallel acceleration [26, 27]. Custom acceleration of algorithms using GPU have become prevalent in high performance computing [28–34]. Secondly, since the algorithm version in [23] is Gaussian noise sensitive, we design a Gaussian noise elimination method that works on the energy histogram directly and enjoys low complexity. Thirdly, we asymptotically analyze the performance limit of the algorithm under high photon activity. The numerical results show that the parallel implementation on GPU efficiently corrects the pileups with a significant speedup ratio exceeding 200. The acceleration ratio remains nearly linear as the problem size scales up. The algorithm also achieves high energy resolution with the existence

of Gaussian noise.

The rest of the paper is organized as follows. In Section II, the pulses pileup problem is described and the original sequential algorithm is presented. In Section III, the Gaussian noise elimination and the parallel implementation is elaborated. The algorithm performance under high activity is asymptotically analyzed. Section IV presents the numerical results and the conclusion is drawn in Section V.

II. PROBLEM STATEMENT

The response of an X-ray or gamma-ray detector to incident photons can be modeled as

$$s[n] = \sum_{k=1}^{\infty} \Phi_k[n - \lceil T_k \rceil], \quad (1)$$

where the arrival times $\{T_k, k \geq 1\}$ are unknown and form a Poisson process with known intensity λ , and Φ_k is the k -th recorded electrical pulse with duration X_k and energy Y_k . We assume that $\{(X_k, Y_k), k \geq 1\}$ is a sequence of independent and identically distributed integer-valued discrete random variables, independent of $\{T_k, k \geq 1\}$, with finite expectation and common probability mass function $p_{X,Y}$. In practice, both X_k and Y_k cannot be observed in experiments. Instead, the pileup's duration X'_k and energy Y'_k can be inferred from the experiments (as described in Fig. 2), and the common probability mass function is denoted by $p_{X',Y'}$.

The pileup correction algorithm described in [23] estimates the marginal distribution of the pulse's energy p_Y , i.e., the histogram of the computed energies, given the distribution of the observed durations and energies of pileups $\{(X'_k, Y'_k)\}$. We recall it for convenience in Algorithm 1. In this algorithm, the joint probability mass function $p_{X',Y'}$ is estimated by a bi-dimensional histogram on $\llbracket 0, N \rrbracket \times \llbracket 0, M \rrbracket$, where N and M are the number of histogram bins for duration and energy, respectively. This algorithm efficiently recovers the pulses' energy distribution from the histogram of the duration and energies of overlapping pulses, i.e., the pileups. However, Algorithm 1 is sequential in nature and it recursively corrects the probability of each energy bin, which is slow when the number of energy bins M is large and the energy resolution is high.

Algorithm 1 Energy domain based pileup correction algorithm proposed by [23].

```

1: INIT:  $y[0][0] \leftarrow 1, y[n][m] \leftarrow 0$  for other  $n, m$ 
2: for  $n = 0, 1, \dots, N$  do
3:   for  $m = 0, 1, \dots, M$  do
      $y[n, m] \leftarrow e^{-\lambda} y[n-1, m] +$ 
      $(1 - e^{-\lambda}) \sum_{k=0}^{n-1} \sum_{l=0}^m y[n-1-k, l] p_{X',Y'}[k, m-l]$ 
4:   end for

```

```

129 5: end for
130 6: for  $n = 0, 1, \dots, N$  do
131 7:   for  $m = 0, 1, \dots, M$  do
132    $k[n, m] \leftarrow \frac{y[n, m]}{\lambda y[n, 0]} - \frac{1}{m} \sum_{k=1}^{m-1} (m-k) \frac{y[n, k]}{y[n, 0]} k[n, m-k]$  (3)
133 8:   end for
134 9: end for
135 10: for  $m = 0, 1, \dots, M$  do
136    $p_Y(m) \leftarrow k[N, m] - k[N-1, m]$ 
137 11: end for

```

Algorithm 1 estimates efficiently the energy histogram from measurements of the duration and energies of overlapping pulses. However, the current sequential execution of the algorithm implies that its performance can be further improved by parallel optimization. In addition, the algorithm is sensitive to Gaussian noise, and the collected signal needs to be pre-processed to effectively filter Gaussian noise.

III. METHODOLOGY

In this section, we first present an energy histogram refinement to eliminate Gaussian noise directly. Then, the multi-core based pileup correction parallelization is presented. Finally, the complexity of the parallel version and the performance under high photon activity are analyzed.

A. Gaussian noise elimination

In a typical nuclear pulse processing procedure, it requires to remove background noise from the signal output of the preamplifier. Filters are essential components that remove unwanted frequencies from signals. High-pass and low-pass filters allow only specific frequency range of signals to pass through while blocking out other frequencies, and are used widely in electronics such as power supply circuits. They help to prevent unwanted frequencies from going through certain components in these systems, thereby minimizing distortion from noise and interference. For Gaussian noise that spreads over a wide range of frequencies, it usually requires Wiener filter, Kalman filter or adaptive filter. However, these filtering algorithms are highly complex and work on the signal sequence directly.

In this work, we propose to eliminate Gaussian noise from the calculated energy histogram instead of the original signal sequence, which significantly reduces the noise filtering complexity. More specifically, the observed signal consists of the detector output corrupted by noise, i.e.,

$$F = I + N, \quad (4)$$

where F , I and N represent the signal sequence observed, ideal signal sequence without noise, and the Gaussian white noise, respectively.

Recall that $p_{X', Y'}[n][m]$ is the estimated bi-dimensional energy distribution histogram on $\llbracket 0, N \rrbracket \times \llbracket 0, M \rrbracket$ from the observed signals F . Let $\hat{p}_{X', Y'}[n][m]$ be the estimated bi-dimensional histogram after Gaussian noise is removed from F . When the SNR is good enough and the duration bins in the histogram are larger than those in the energy, the Gaussian noise has little effect on the time dimension. When the observation is sufficient, we have

$$\hat{p}_{X', Y'}(X' = n | Y' = m) \approx p_{X', Y'}(X' = n | Y' = m) \quad (5)$$

According to the related theorems of probability theory, conditional probability can be expressed by the following formula

$$\frac{\hat{p}_{X', Y'}[n][m]}{p_{X', Y'}[n][m]} = \frac{\hat{p}_{X', Y'}(X' = n | Y' = m) p_{Y'}(Y' = m)}{p_{X', Y'}(X' = n | Y' = m) p_{Y'}(Y' = m)} \quad (6)$$

$p_{Y'}(Y' = m)$ is the probability that the energy is m in the original signal statistics and $\hat{p}_{Y'}(Y' = m)$ is the probability that the energy is m in the original signal statistics after removing the noise. Their values can be obtained from the energy spectrum. Set f_I and f_F are the distribution of I and F , respectively. Therefore, the energy distribution histogram of ideal signals can be estimated by

$$\hat{p}_{X', Y'}[n][m] \approx p_{X', Y'}[n][m] \frac{f_I(Y' = m)}{f_F(Y' = m)}. \quad (7)$$

The distribution functions f_F can be empirically estimated from the original signal sequence. The distribution function f_I of the ideal signal sequence without noise can be estimated as the follows. Let Φ_F , Φ_I and Φ_N be the characteristic functions of F , I and N . From (4), we have

$$\Phi_F(u) = \Phi_I(u) \Phi_N(u). \quad (8)$$

The Target signal is the gamma signals and the characteristic function of noise converges to 0 slowly in the case of low noise, by rearranging the terms in (8), we have

$$\Phi_I(u) = \frac{\Phi_F(u)}{\Phi_N(u)} \quad (9)$$

Ideally, f_I is recovered by taking a Fourier transform

$$f_I(x) = \int_{-\infty}^{\infty} e^{-i2\pi ux} \Phi_I(u) du. \quad (10)$$

The characteristic functions Φ_F and Φ_N can be estimated by the original signal sequence and the pre-collected noise signals without a radioactive source present, respectively.

This Gaussian noise elimination algorithm only calculates the statistical characteristics of the signal, which avoids the difficulty of the traditional filtering algorithm working directly on the original signal sequence and thus has high complexity, and significantly reduces the computational complexity while still maintaining the same noise elimination ability.

B. Parallel implementation

It is noticed that the computation of (2) is the most time consuming step of Algorithm 1. As shown in Fig. 3, the computation of $y[n][m]$ involves all $y[k][l]$ for $k < n, l \leq m$. In the sequential implementation of Algorithm 1, each element of the matrix y is calculated one by one. Therefore the algorithm complexity depends on the size of matrix y , and as the size grows, the overall algorithm is slow. Fortunately, the accumulation term $\sum_{k=0}^{n-1} \sum_{l=0}^m y[n-1-k, l] p_{X', Y'}[k, m-l]$ can be parallelized in a many-core GPU environment and the execution speed is expected to increase.

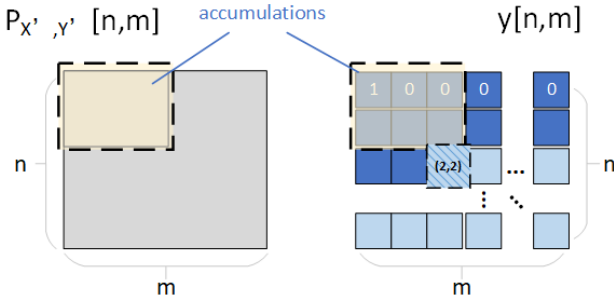


Fig. 3. CPU-based implementation of equation 2 computation.

We describe the parallel implementation of pileup correction in Algorithm 2. There are two challenges faced by the migration of pileup correction to a GPU environment. First, better parallelism can be achieved by element-wise scheduling for matrix y . The idea is to assign an individual GPU core to each element of matrix y so that the accumulation can be computed simultaneously. Fig. 4 illustrates how parallelization is achieved by element-wise scheduling. For an $N \times M$ matrix y , NM GPU cores are required and N steps are executed in order to compute matrix y . At step i , the GPU cores that assigned to elements with row index $n, \dots, N-1$ calculate its accumulative as in (2) with partial information obtained from step $i-1$. Also, the GPU cores assigned to elements with row index n will finish the computation of (2) and their results will be accessed at step $i+1$.

The specific implementation steps are as follows:

1. Assign the first row of the matrix.
2. Implement the GPU core calculation method: for the $[i, j]$ element $y[i, j]$, the value V on the core $[idx, idy]$ is overwritten as $V = V + y[i, j] * P[idx-i-1, idy-j]$
3. Traverse through $N \times M$ elements, performing column traversal for each row.
4. Complete the generation of the target matrix.

As shown in Fig. 4, for an $N \times M$ matrix y , $N \times M$ GPU cores are required and N steps are executed in order to compute matrix y . At step i , the GPU cores that assigned to elements with row index $n, \dots, N-1$ calculate its accumulative as in (2) with partial information obtained from step $i-1$. Also,

after the GPU cores assigned to elements with row index n finish the computation of (2), their results can be accessed at step $i+1$. The pseudo-code of the algorithm is summarized in Algorithm 2.

Algorithm 2 Parallel implementation.

```

1: ##Denoising p[][]
2: for j = 1, ... M do
    p(j) ← ∑_{i=1}^N p(i, j)
    N(j) ← ∫_i^{i+1} N(μ, σ^2)(N+1 = ∞)
3: end for
    cufft { fftp[] = cufft(p[])
           fftN[] = cufft(N[])
           fftp'[] ← fftp[]/fftN[]
           p'[] ← ifft(fftp'[])
4: for i = 1 ... N do
5:   for j = 1 ... M do
6:     p[i][j] ← p[i][j]*p'[j]
7:   end for
8:   for each GPU core P_{n,m} do
9:     if n == 0, m == 0 then
10:      y[0][0] ← 1
11:     else
12:      y[n][m] ← 0
13:     end if
14:   end for
15:   ##Compute y[][]
16:   for i = 1, ... N do
17:     for each GPU core P_{n,m} do
18:       if n ≥ i then
19:         y[n, m] ← y[n, m] + (1-e^{-λ}) ∑_{l=0}^m y[i-1, l] p_{X', Y'}[n-i, m-l]
20:       end if
21:       if n == i then
22:         y[n, m] ← y[n, m] + e^{-λ} y[i-1, m]
23:       end if
24:     end for
25:   end for
26:   ##Compute k[][]
27:   for each GPU core P_{n,m} do
28:     k[n, m] ← y[n, m] / λ y[n, 0]
29:   end for
30:   for j = 1, ... M do
31:     for each GPU core P_{n,m} do
32:       if m > j then
33:         k[n, m] ← k[n, m] - j/m k[n, j] y[n, m-j] / y[n, 0]

```

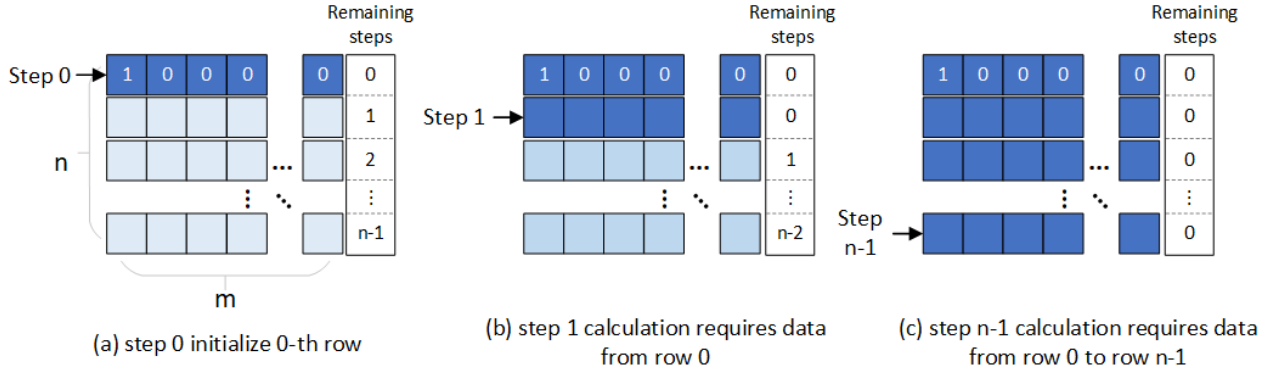


Fig. 4. GPU-based implementation of equation 2 computation.

```

303 28:     end if
304 29:     if n == N then
305
306          $p_Y(m) \leftarrow k[n, m] - k[n-1, m]$ 
307
308     end if
309 end for

```

C. Time Complexity

For the original implementation of Algorithm 1, the nested loop for calculating the matrix y (Line 2 to Line 5) dominates the time complexity. In the loop, the calculation of $y[n, m]$ as in (2) involves an accumulation operation of time complexity $O(NM)$, and thus the complexity of the nested loop is $O(N^2M^2)$. Therefore, the overall time complexity of Algorithm 1 is $O(N^2M^2)$.

Regarding the parallel implementation of Algorithm 2, for each GPU core, it first goes through a loop of N steps (Line 14 to Line 21). At each step, each core runs an accumulation operation of complexity $O(M)$. Secondly, each GPU core goes through a loop of M steps (Line 25 to Line 32), and each step involves a constant number of operations. Therefore, there are NM GPU cores, and for each GPU core, its time complexity is $O(NM)$.

D. Performance limit under high activity

The signal processing approaches in the time-domain are limited when the activity of the radioactive source increases too much [22]. On the other hand, the statistical approach in the energy-domain is also limited under high activity. Naturally, as the photon activity continues to increase, the number of pulses contained in the pileup also increases, and the corresponding energy of the pileup also increases. Recall that we assume that $\{Y'_k, k \geq 1\}$ is the sequence of the pileups' energy, and $p_{Y'}$ is its probability mass function. We can conclude the following theorem.

Theorem 1. For $\forall \epsilon > 0$, there exists some value $M > 0$, when the activity $\lambda \geq N$, the probability that the accumulated energy of any pileup is less than M satisfies

$$p_{Y'}(Y' \leq M) < \epsilon, \quad (12)$$

where

$$N = \begin{cases} \max(\frac{2M^2\pi}{\epsilon^2}, \ln \frac{1}{\epsilon}) & e_0 \geq 1 \\ \max(\frac{2M^2\pi}{\epsilon^2 e_0^2}, \ln \frac{1}{\epsilon}) & e_0 < 1 \end{cases}$$

and e_0 is energy resolution.

The detailed proof can be found in Appendix V. According to Theorem 1, if the desired energy resolution e_0 is fixed, there exists an N , when the activity λ is greater than N , the probability that the pileup energy is bounded approaches to 0. In other words, when the activity is too high, a large number of pulses stack together, so that the accumulated energy of pileup grows to infinity. Therefore, the energy domain algorithm is limited as well.

IV. PERFORMANCE EVALUATION

In this section, we present performance evaluation of Algorithm 2 both on simulated data and real data collected from the Shanghai Synchrotron Radiation Facility (SSRF). Throughout the experiments, Algorithm 1 is evaluated with an Intel(R) Xeon(R) Gold 6230 CPU, and Algorithm 2 is evaluated on the same platform with an additional NVIDIA Tesla V100S PCIe 32GB card.

A. Results Based on Simulations

The ideal density used in [23] was used for generating simulated pileup durations and energies. It consists of a mixture of six Gaussian and one gamma distribution to simulate the Compton background. The probability density of the mixture f_Y is proportional to

$$0.5g + 10N(40, 1) + 10N(112, 1) + N(50, 2) + N(63, 1) + 2N(140, 1) + N(200, 1),$$

where $N(\mu, \sigma^2)$ is the probability density of a normal distribution with mean μ and variance σ^2 . An additional Gaussian noise $N(0, 4)$ was added to the simulated energies to verify the effectiveness of proposed Gaussian noise elimination method. The probability density of the gamma distribution is given by $g(x) = (0.5 + x/200)e^{-(0.5+x/200)}$ for $x \geq 0$.

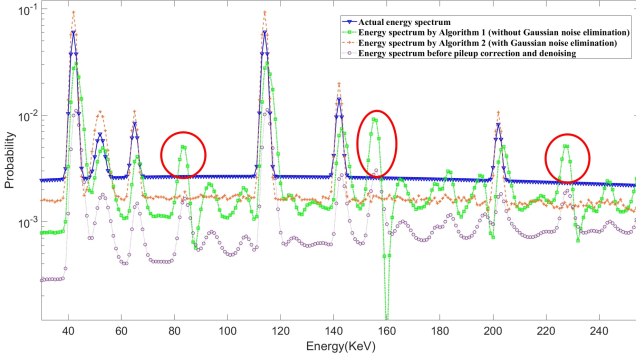


Fig. 5. Pileup correction by Algorithm 2. $\lambda = 2,000,000$ photons per second. $N = 16$, $M = 256$.

Fig. 5 shows that the energy spectrum is efficiently corrected by parallel Algorithm 2: all the fake peaks (at 80 KeV, 152 KeV and 224 KeV) are discarded, and the obtained estimation of the energy spectrum almost perfectly matches the objective density even with a very high photon activity of $\lambda = 2,000,000$. Compared to the original algorithm without the Gaussian noise elimination in Algorithm 1, the Gaussian noise elimination method helps to remove the fake peaks caused by Gaussian noise and improve the peak-to-background-ratio in the resulted spectrum.

Next, we demonstrate the performance of Algorithm 2 in terms of execution speed. The speedup ratio ρ is defined in (13), where t_1 and t_2 are the run time of Algorithm 1 and Algorithm 2, respectively.

$$\rho = t_1/t_2. \quad (13)$$

Table 1 presents the run time of Algorithm 1 and Algorithm 2 with different combination of energy bins and duration bins. With duration bin $N = 128$ and energy bin $M = 2048$, the speedup ratio exceeds 200. In Fig. 6, the speedup ratio ρ is illustrated. The results in Fig. 6 demonstrates a nearly linear speedup as the numbers of duration and energy bins N , M increase. It is obvious that at the current problem scale, the NVidia V100 has sufficient cores to maximize the parallelism and the execution speed of the pileup correction has been greatly improved.

B. Results Based on Real Data

We obtained the real signal dataset from SSRF. The radioactive source is the element Cu-63. The detector used is Vortex[®] EX90 Silicon Drift Detector.

As shown in the figure 7, the duration of the real data is minimally affected by the energy magnitude. With sufficiently good SNR, the noise reduction algorithm is valid.

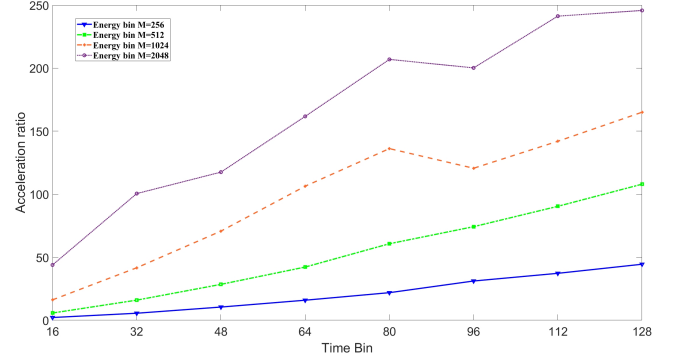


Fig. 6. Speedup ratio ρ

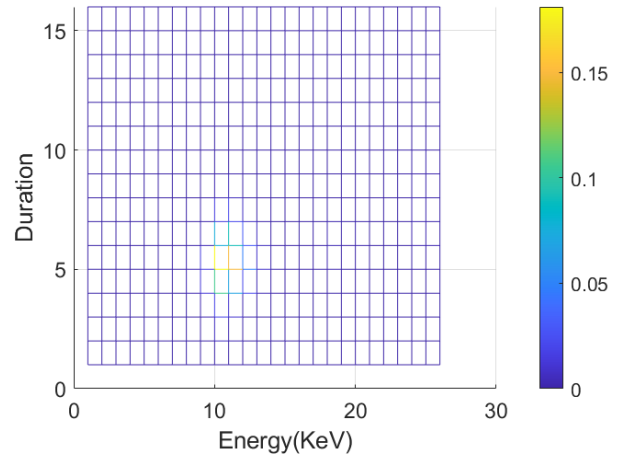


Fig. 7. Joint probability distribution of duration and energy magnitude

In Fig. 8, we set energy bin $M = 256$ and the activity rate λ at $1.62E+5$ photons per second. The results reveal that the algorithm effectively eliminates the pseudo-peak (highlighted in the red circle) induced by the stacking effect, underscoring its substantial correction effect on real data. Moreover, the successful implementation of the algorithm on real data en-

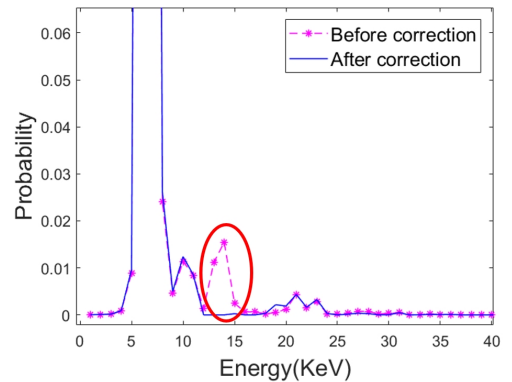


Fig. 8. Pileup correction by Algorithm 2 on real data from SSRF.

Table 1. Running Time (ms)

#duration bins \ #energy bins	Algorithm 1				Algorithm 2			
	256	512	1024	2048	256	512	1024	2048
16	45	143	533	2104	21	25	33	48
32	133	510	1956	7836	24	32	47	78
48	282	1081	4312	17271	28	38	61	147
64	492	1898	7760	30399	31	45	73	188
80	743	3094	11844	48001	34	51	87	232
96	1120	4229	16998	68054	36	58	141	340
112	1451	5783	24323	92146	39	64	165	382
128	1865	7554	30524	119913	42	70	185	488

hances the feasibility of its industrial application and broad-
ens its potential applications. This demonstrates that the algo-
rithm transcends theoretical implications, offering significant
practical value at the industrial level.

C. Evaluation of High Activity Rate

Theorem 1 describes the relationship between the energy
resolution e_0 , the activity rate λ and the upbound of the ac-
cumulated pileup energy M . We evaluate this relationship in
Fig. 9. We set the energy resolution $e_0 = 1\text{KeV}$ and the
accumulated energy bound $M = 3000\text{KeV}$ according to the
Ortec GEM-S detector. Fig. 9 describes the probability that
the accumulated pileup energy Y' is not greater than the en-
ergy bound $M = 3000\text{KeV}$ with different activity rate λ , un-
der the fixed energy resolution $e_0 = 1\text{KeV}$. We also develop
a Monte-Carlo simulation of pulse pile-up effect in gamma
spectrum to simulate the performance limit under high activ-
ity, which is presented in Fig. 10. The simulation results and
the theoretical results show a similar probability decay trend.
Although the upper limit of the activity obtained by the the-
oretical analysis is higher than that of the simulated one, this
theoretical analysis still has guiding significance in practice.

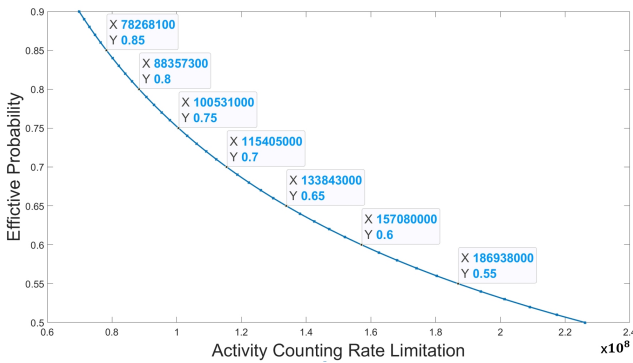


Fig. 9. Theoretical performance limit under high activity rate

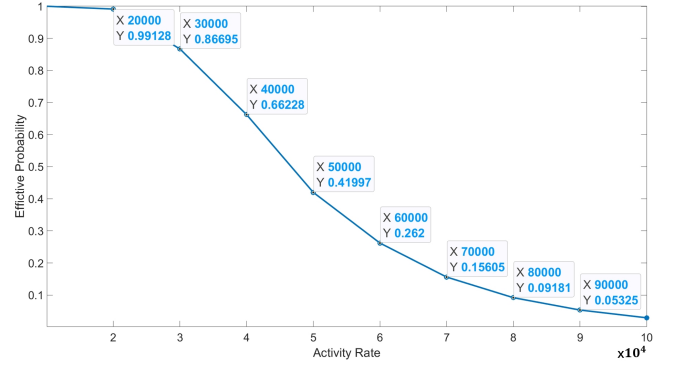


Fig. 10. Simulated performance limit under high activity rate

V. CONCLUSION

The pileup correction algorithm derives an analytical rela-
tion between the probability mass function of the observed
pileups and the probability mass function of the pulses. The
algorithm can efficiently correct the pileup effect, but is very
slow in practice when the energy resolution is high. In this pa-
per, we firstly proposed to accelerate the pileup correction by
GPU coprocessors. We illustrated the difficulties and the so-
lutions encountered when deploying GPU technology to per-
form pileup correction. It is also noted that collected sig-
nals often contain significant Gaussian noise that poses chal-
lenges for correction. We secondly introduced a Gaussian
noise elimination method that directly remove Gaussian noise
from the calculated energy histogram, which significantly re-
duces the noise filtering complexity. Thirdly, we theoretically
analyzed the performance limit of the algorithm under high
activity. The analysis shows that when the photon activity is
high, there exists an energy threshold such that the accumu-
lated pileup energy is no more than this energy threshold with
nearly zero probability. Finally, we evaluated the algorithms
with an NVIDIA Tesla V100S PCIe 32GB, which showed
that the pileup correction has achieved a linear speedup on the
GPU. The overall performance of the GPU-accelerated pileup
correction method can outperform its CPU counterpart by a
significantly large factor, more than 200, which is important
in real-time processing nuclear spectrometric data in practice.
It also achieves a better energy resolution in the existence of
Gaussian compared with the original plain approach. Further-

more, by applying the algorithm to real signals collected from SSRF, we successfully demonstrated its effectiveness and potential for industrial applications.

APPENDIX

A. Proof For Theorem 1

Proof. In our previous introduction we could model pulse stacking as a compound Poisson process with rate λ . In this section, constraints of the algorithm will be outlined to limit the count rate λ of the radioactive source. This is done to ensure that the energy value of the pulse cluster does not become excessively large. We assume that the stochastic process of the original pulse is A, the compound Poisson process is Y. It was proposed that

$$\Phi_Y(u) = e^{-\lambda} e^{\lambda \Phi_A(u)}$$

$\hat{\Phi}_X$ is the characteristic function of X [35].

When the count rate of pulses is too high, the spectrometer will not return to steady state. For any given energy M, the probability that the piled energy is less than M approaches 0

Because we're looking at the lower bound, so e_0 here represents the resolution of the nuclear spectrometer

$$\Phi_A(u) = e^{iue_0}$$

$\hat{f}_Y[k]$ is the probability that the energy from the pile is between k and k + 1, the following theorem is established:

$$\begin{aligned} \hat{f}_Y[k] &= \frac{1}{2M} \sum_{m=-M}^{m=M-1} \Phi_Y\left[\frac{m\pi}{M}\right] e^{-\frac{i\pi mk}{M}} \\ &= \frac{1}{2M} \sum_{m=-M}^{m=M-1} e^{\lambda} e^{\lambda e^{i\frac{m\pi}{M}e_0}} e^{-\frac{i\pi mk}{M}} \\ &= \frac{1}{2M} \sum_{m=-M}^{m=M-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0) + \lambda i \sin(\frac{m\pi}{M}e_0)} e^{-\frac{i\pi mk}{M}} \\ &= \frac{1}{2M} \sum_{m=-M}^{m=M-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} e^{i(-\frac{\pi mk}{M} + \lambda \sin(\frac{m\pi}{M}e_0))} \end{aligned}$$

The target value, representing the probability of a specific energy, is a real number. Therefore, the imaginary part is not considered in the Fourier transform.

$$\begin{aligned} LHS &= \frac{1}{2M} \sum_{m=-M}^{m=M-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} \cos\left(\lambda \sin\left(\frac{m\pi}{M}e_0\right) - \frac{\pi mk}{M}\right) \\ &\quad + \frac{1}{2M} \sum_{m=1}^{m=M-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} \cos\left(\lambda \sin\left(\frac{m\pi}{M}e_0\right) - \frac{\pi mk}{M}\right) \\ &\quad + \frac{1}{2M} \sum_{m=-M}^{m=-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} \cos\left(\lambda \sin\left(\frac{m\pi}{M}e_0\right) - \frac{\pi mk}{M}\right) \\ &\quad + \frac{1}{2M} e^{(-\lambda)} \end{aligned}$$

We are considering the probability for pile-up, so if $k < 0$, $\hat{f}_Y[k] = 0$ we consider the sum of pro from $k = 1$ to $k = M-1$, and for the sake of clarity, we establish the following conventions:

$$C = \frac{1}{2} e^{(-\lambda)} \quad (14)$$

$$g(k) = \lambda \sin\left(\frac{m\pi}{M}e_0\right) - \frac{\pi mk}{M} \quad (15)$$

By substituting Eq 14 and Eq 15 into the original equation, the equation can be simplified, allowing us to continue the derivation.

$$\begin{aligned} \sum_{k=1}^{k=M-1} \hat{f}_Y[k] &\leq \frac{1}{2M} \sum_{k=1}^{k=M-1} \sum_{m \neq 0} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} * \cos(g(k)) + C \\ &\leq \frac{1}{2M} \sum_{m \neq 0} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} \left(\sum_1^{M-1} \cos(g(k)) \right) + C \\ &\leq \frac{1}{2M} \sum_{m \neq 0} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} * M + C \\ &\leq \sum_{m \neq 0} \frac{1}{2} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} + C \end{aligned}$$

Since the symmetry of the cosine function

$$LHS \leq \sum_{m=1}^{m=M-1} e^{-\lambda + \lambda \cos(\frac{m\pi}{M}e_0)} + C$$

Through the application of the double angle formula of trigonometric functions and the fundamental properties of elementary functions, we obtain:

$$\cos t - 1 = -2 \sin^2 \frac{t}{2} \quad (16)$$

when $0 < t < \pi$

$$\sin \frac{t}{2} > \frac{t}{\pi} \quad (17)$$

so when $e_0 > 1$, let $t = \frac{m\pi}{M}$ and substitute Eq 16 and Eq 17 into the expression:

$$\begin{aligned} LHS &\leq \sum_{m=1}^{m=M-1} e^{-2\lambda(\frac{m}{M})^2} + C \\ &\leq \int_0^{M-1} e^{-(\frac{\sqrt{2\lambda}}{M}m)^2} dm + C \\ &\leq \frac{M}{\sqrt{2\lambda}} \int e^{(-t^2)} dt + C \\ &\leq M \sqrt{\frac{\pi}{2\lambda}} + C \end{aligned}$$

For $M > 0$

$$N = \max\left(\frac{2M^2\pi}{\epsilon^2}, \ln \frac{1}{\epsilon}\right)$$

Now consider the other side, when $e_0 < 1$, do the same thing
as above, except $t = \frac{m\pi}{M}e_0$

When $\lambda > N$

$$LHS \leq \frac{M}{e_0} \sqrt{\frac{\pi}{2\lambda}} + C$$

For $M > 0$

$$N = \max\left(\frac{2M^2\pi}{\epsilon^2 e_0^2}, \ln \frac{1}{\epsilon}\right)$$

522

□

The following results can be obtained from summarizing the
above steps:

$$N = \begin{cases} \max\left(\frac{2M^2\pi}{\epsilon^2}, \ln \frac{1}{\epsilon}\right) & e_0 \geq 1 \\ \max\left(\frac{2M^2\pi}{\epsilon^2 e_0^2}, \ln \frac{1}{\epsilon}\right) & e_0 < 1 \end{cases}$$

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